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pvcracks: Open tools for photovoltaic crack analysis and electrical simulation

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Summary

pvcracks is an open-source Python package that links photovoltaic (PV) electroluminescence (EL) images with electrical reliability analysis. It segments cracks, dark regions, busbars, and non-cell regions in cell EL images; derives compact image representations; estimates image-associated power loss; and builds cell-resolved electrical simulations. The `pvspice` subpackage writes one-diode SPICE circuits and supports cell, module, and string simulations with spatially varying photocurrent, series resistance, and shunt resistance. The package, documentation, trained artefacts, and reproducible examples are available for research and engineering use (Hartley et al., 2024).

Statement of need

EL imaging is widely used to identify manufacturing defects and in-service damage in PV modules. An EL pattern, however, does not by itself specify the electrical consequence of a defect. A crack may affect a cell differently depending on its location, the cell interconnection, operating conditions, and the associated changes in current or resistance. Analysts therefore need to move from image-derived features to cell parameters and then to a module or string current-voltage (IV) curve. This work is often divided among image scripts, manually edited circuit netlists, and proprietary tools. The resulting workflow can be difficult to inspect, reproduce, or adapt.

pvcracks provides a documented path from cell EL images to feature segmentation and data-driven power-loss estimates. It also provides physically interpretable electrical simulation when cell parameters or degradation maps are available. The package is intended for PV reliability researchers, diagnostic-tool developers, and engineers studying cracking, reverse bias, partial shading, and potential-induced degradation (PID). The workflows use file-backed, parameterized inputs so users can replace the included images, IV data, and module layouts with their own data.

State of the field

The open PV software ecosystem includes general system modelling in pvlib-python (Holmgren et al., 2018), EL image processing in PVimage (Pierce & others, 2020), degradation analysis in RdTools (NREL, 2024), and circuit-based mismatch simulations in PVMismatch (Corporation, 2024). pvcracks is designed to work alongside these packages. Its specific contribution is the connection between EL-derived feature information and cell-resolved electrical simulations.

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pvspice accepts maps of photocurrent, series resistance, and shunt resistance. It can therefore represent spatially heterogeneous damage within a module and across a string.

Software design

The default pvcracks distribution includes the documented analysis tools, pretrained inference, XGBoost workflow, and pvspice without Ray or experiment-management dependencies. Model retraining and hyperparameter optimization are supplied through the separately installable pvcracks-training companion distribution. `MultiSolSegment` produces overlapping masks for cracks, dark regions, busbars, and non-cell areas in multi-busbar EL images. Sanghi *et al.* describe the segmentation method and benchmarks (Sanghi et al., 2026). The labelled images and masks (DataHub, 2025b) and model weights (DataHub, 2025a) are cited separately so users can retrieve the exact data and trained artefacts.

The package also includes a convolutional variational autoencoder for compact image representations used in triage and clustering. A separate XGBoost workflow relates processed image features to maximum-power loss, following the crack-power-loss study of Braid *et al.* (Braid et al., 2025). The pretrained VAE is archived separately (DataHub, 2025c). These models support inspection and inference; the training distribution is required only when users need to retrain or tune models.

pvspice creates inspectable one-diode circuit netlists, runs a compatible local SPICE engine, and parses IV and EL outputs. Its API separates netlist generation, simulation execution, raw-output parsing, and degradation-map construction. It supports series and half-cut module topologies, bypass diodes, and full strings. Users can inspect the electrical model, change cell-level parameters, and run the same workflow with a compatible simulator.

The documentation provides focused examples rather than using this paper to report application-specific plots. `example_pvspice_reverse.ipynb` demonstrates reverse-bias and mismatch analysis. `example_pvspice_thermal.ipynb` applies the workflow to UC Merced partial-shading IV and temperature data. The experimental context is documented in PVRW proceedings (Photovoltaic Reliability Workshop, 2026) and by Islam and Kurtz (Islam & Kurtz, 2026). The mini-module example uses cell-level parameter and EL data from a PVSC cracked-cell study (Prado Santamaria et al., 2021). `example_pvspice_PID_pingel.ipynb` applies spatial shunt-resistance patterns from Pingel *et al.* (Pingel et al., 2010) to a module string. The examples provide reproducible starting points for new module layouts and degradation maps. The synthetic EL-IV workflow was also presented at EU PVSEC (Jost et al., 2024).

Research impact statement

pvcracks makes image processing, learned estimation, and cell-resolved circuit analysis available as inspectable Python code. Tests, examples, documentation, trained weights, and data references give users clear entry points for reproducing and extending a workflow. The project is distributed through the Department of Energy's DuraMAT DataHub and has a citable software release (Hartley et al., 2024). This paper cites the software, data, trained artefacts, and prior PV studies at their points of use so users can identify the provenance of each component.

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AI usage disclosure

OpenAI Codex (GPT-5) assisted with code refactoring, test scaffolding and documentation updates. The authors wrote the initial code and made architectural and design decisions; they reviewed, edited, and validated all AI-assisted content and retain responsibility for the submitted work.

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